



MISCELLANEOUS COMPONENTS

We often ignore them, but your PC's cabinet, sound system and input devices play a significant role in your PC's speed and stability

While your processor, graphics card and hard disk steal the limelight in terms of technology and performance, there are other components that play a role in boosting performance but are usually ignored—such as

your system cabinet, input devices, speakers and even the power supply. These components won't affect your frame rates or the time taken by Photoshop to start up, but they do facilitate smoother work and allow your computer to be upgraded with

newer and more powerful 'centre stage' devices. Let's take a look at some of these devices, peripherals and external expansion interfaces and understand how they can affect your system's performance and upgradability.

External Bus

With the explosion in the popularity of devices such as scanners, CD-Writers, digital cameras and the like, the need arose for an umbilical cord that would connect them to your computer, speedily and reliably. Due to their relatively low speeds and virtually non-existent expandability, serial, parallel and game ports could not keep pace with the onslaught of new devices that were introduced for home and office computers. Newer standards that would allow these devices to be connected to the computer at high speeds and yet leave room for expansion were needed.

The answer came in the form of USB and FireWire—both these plug-and-play technologies were born to create an external expansion route for newer portable devices and peripherals. While devices such as keyboards and mice do not really require this increased bandwidth, other devices such as scanners, printers, MP3 players, external storage devices and CD-Writers place greater demands on the speed of the port. For example, in its present version, USB 2.0

allows for data transfer speeds of up to 480 Mbps while FireWire allows a speed of 400 Mbps—both are ideally suited to bandwidth-hungry devices such as scanners and digital cameras. Similarly, serial ports have a paltry transfer rate of just 115 Kbps, while parallel ports are marginally better with a transfer rate of 1.5 Mbps.

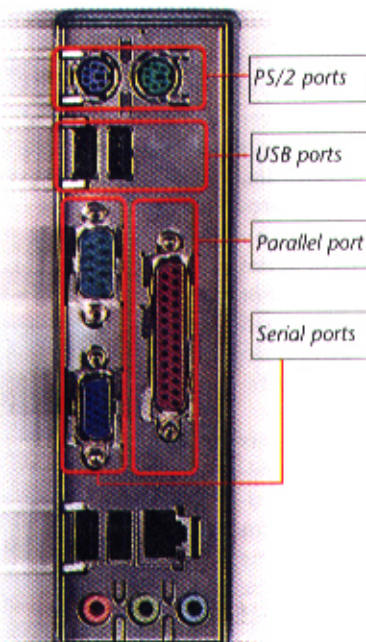
Sometimes, a simple little device such as a PS/2 mouse could be a cause for heartache when, after you've bought it, you discover that your old Baby AT format motherboard does not even have a PS/2 port! Even today, if the temptation to buy one of those new PDAs is just too high, you should check if your computer is equipped with the interface needed for the device. Many older computers have only serial and parallel connectors

and no USB connectors. Therefore, while you could still use a PDA on the serial port, the paltry 112 Kbps maximum transfer rate will leave you tearing your hair out each time you try to synchronise your mail with your PDA or download images from your digital camera. Interfaces such as USB also alleviate expansion problems since they allow devices to be daisy-chained to one another—USB allows up to 128 devices

to be connected this way to a single port on your computer!

When it comes to new interfaces, you also need to ensure that your OS supports them. While recent iterations of Windows (Windows 98 SE onwards) do have inherent support for USB devices, make sure that there are patches or updates available for newer technologies such as USB 2.0. This would be especially relevant with operating systems such as Linux where you would require patches and driver support for these interfaces.

Some of the newer motherboards also feature support for new-generation wireless interfaces for portable devices such as Bluetooth, Infrared and WiFi. If you have an especially generous relative abroad who has sent you a Nokia 7650, ensure that your computer can support it either inherently or through expansion cards if you want to utilise the full potential of the device.



Tips & Tweaks

- Make sure that your serial ports are set to operate at full speed from within Windows. Set the transfer rate to 57600 bits per second.
- Using a PS/2 or USB mouse will give you much smoother control, as the sampling rate of these ports is much higher than that of a serial port.
- If your USB device does not work even after checking connections and compatibility, enable the setting 'Assign IRQ to USB' from the BIOS.